

T 68. Slopes on the graph of the cotangent function Graph $y = \cot x$ and its derivative together for $0 < x < \pi$. Does the graph of the cotangent function appear to have a smallest slope? A largest slope? Is the slope ever positive? Give reasons for your answers.

T 69. Exploring $(\sin kx)/x$ Graph $y = (\sin x)/x$, $y = (\sin 2x)/x$, and $y = (\sin 4x)/x$ together over the interval $-2 \leq x \leq 2$. Where does each graph appear to cross the y -axis? Do the graphs really intersect the axis? What would you expect the graphs of $y = (\sin 5x)/x$ and $y = (\sin(-3x))/x$ to do as $x \rightarrow 0$? Why? What about the graph of $y = (\sin kx)/x$ for other values of k ? Give reasons for your answers.

T 70. Radians versus degrees: degree mode derivatives What happens to the derivatives of $\sin x$ and $\cos x$ if x is measured in degrees instead of radians? To find out, take the following steps.

- a. With your graphing calculator or computer grapher in *degree mode*, graph

$$f(h) = \frac{\sin h}{h}$$

and estimate $\lim_{h \rightarrow 0} f(h)$. Compare your estimate with $\pi/180$. Is there any reason to believe the limit *should* be $\pi/180$?

- b. With your grapher still in degree mode, estimate

$$\lim_{h \rightarrow 0} \frac{\cos h - 1}{h}.$$

- c. Now go back to the derivation of the formula for the derivative of $\sin x$ in the text and carry out the steps of the derivation using degree-mode limits. What formula do you obtain for the derivative?
- d. Work through the derivation of the formula for the derivative of $\cos x$ using degree-mode limits. What formula do you obtain for the derivative?
- e. The disadvantages of the degree-mode formulas become apparent as you start taking derivatives of higher order. Try it. What are the second and third degree-mode derivatives of $\sin x$ and $\cos x$?

3.6

The Chain Rule

How do we differentiate $F(x) = \sin(x^2 - 4)$? This function is the composite $f \circ g$ of two functions $y = f(u) = \sin u$ and $u = g(x) = x^2 - 4$ that we know how to differentiate. The answer, given by the *Chain Rule*, says that the derivative is the product of the derivatives of f and g . We develop the rule in this section.

Derivative of a Composite Function

The function $y = \frac{3}{2}x = \frac{1}{2}(3x)$ is the composite of the functions $y = \frac{1}{2}u$ and $u = 3x$.

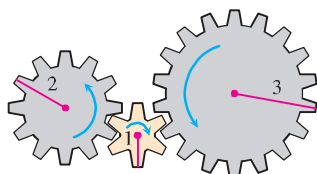
We have

$$\frac{dy}{dx} = \frac{3}{2}, \quad \frac{dy}{du} = \frac{1}{2}, \quad \text{and} \quad \frac{du}{dx} = 3.$$

Since $\frac{3}{2} = \frac{1}{2} \cdot 3$, we see in this case that

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

If we think of the derivative as a rate of change, our intuition allows us to see that this relationship is reasonable. If $y = f(u)$ changes half as fast as u and $u = g(x)$ changes three times as fast as x , then we expect y to change $3/2$ times as fast as x . This effect is much like that of a multiple gear train (Figure 3.23). Let's look at another example.



C: y turns B: u turns A: x turns

FIGURE 3.23 When gear A makes x turns, gear B makes u turns and gear C makes y turns. By comparing circumferences or counting teeth, we see that $y = u/2$ (C turns one-half turn for each B turn) and $u = 3x$ (B turns three times for A's one), so $y = 3x/2$. Thus, $dy/dx = 3/2 = (1/2)(3) = (dy/du)(du/dx)$.

EXAMPLE 1 The function

$$y = (3x^2 + 1)^2$$

is the composite of $y = f(u) = u^2$ and $u = g(x) = 3x^2 + 1$. Calculating derivatives, we see that

$$\begin{aligned}\frac{dy}{du} \cdot \frac{du}{dx} &= 2u \cdot 6x \\ &= 2(3x^2 + 1) \cdot 6x \\ &= 36x^3 + 12x.\end{aligned}$$

Calculating the derivative from the expanded formula $(3x^2 + 1)^2 = 9x^4 + 6x^2 + 1$ gives the same result:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(9x^4 + 6x^2 + 1) \\ &= 36x^3 + 12x.\end{aligned}$$

The derivative of the composite function $f(g(x))$ at x is the derivative of f at $g(x)$ times the derivative of g at x . This is known as the Chain Rule (Figure 3.24).

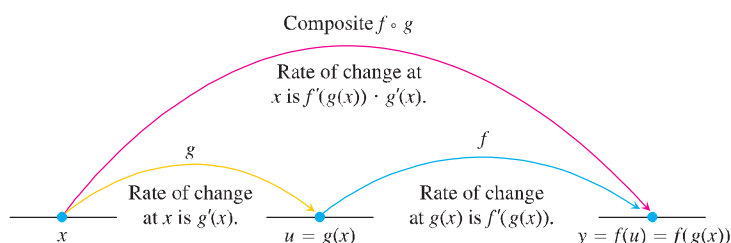


FIGURE 3.24 Rates of change multiply: The derivative of $f \circ g$ at x is the derivative of f at $g(x)$ times the derivative of g at x .

THEOREM 2—The Chain Rule If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x , and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

where dy/du is evaluated at $u = g(x)$.

Intuitive “Proof” of the Chain Rule:

Let Δu be the change in u when x changes by Δx , so that

$$\Delta u = g(x + \Delta x) - g(x).$$

Then the corresponding change in y is

$$\Delta y = f(u + \Delta u) - f(u).$$

If $\Delta u \neq 0$, we can write the fraction $\Delta y/\Delta x$ as the product

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \quad (1)$$

and take the limit as $\Delta x \rightarrow 0$:

$$\begin{aligned}
 \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \\
 &= \lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} \cdot \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} && \text{(Note that } \Delta u \rightarrow 0 \text{ as } \Delta x \rightarrow 0 \\
 &= \frac{dy}{du} \cdot \frac{du}{dx}. && \text{since } g \text{ is continuous.)}
 \end{aligned}$$

The problem with this argument is that it could be true that $\Delta u = 0$ even when $\Delta x \neq 0$, so the cancellation of Δu in Equation (1) would be invalid. A proof requires a different approach that avoids this flaw, and we give one such proof in Section 3.9. ■

EXAMPLE 2 An object moves along the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = \cos(t^2 + 1)$. Find the velocity of the object as a function of t .

Solution We know that the velocity is dx/dt . In this instance, x is a composite function: $x = \cos(u)$ and $u = t^2 + 1$. We have

$$\begin{aligned}
 \frac{dx}{du} &= -\sin(u) && x = \cos(u) \\
 \frac{du}{dt} &= 2t. && u = t^2 + 1
 \end{aligned}$$

By the Chain Rule,

$$\begin{aligned}
 \frac{dx}{dt} &= \frac{dx}{du} \cdot \frac{du}{dt} \\
 &= -\sin(u) \cdot 2t && \frac{dx}{du} \text{ evaluated at } u \\
 &= -\sin(t^2 + 1) \cdot 2t \\
 &= -2t \sin(t^2 + 1).
 \end{aligned}$$

“Outside-Inside” Rule

A difficulty with the Leibniz notation is that it doesn’t state specifically where the derivatives in the Chain Rule are supposed to be evaluated. So it sometimes helps to think about the Chain Rule using functional notation. If $y = f(g(x))$, then

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x).$$

In words, differentiate the “outside” function f and evaluate it at the “inside” function $g(x)$ left alone; then multiply by the derivative of the “inside function.”

EXAMPLE 3 Differentiate $\sin(x^2 + x)$ with respect to x .

Solution We apply the Chain Rule directly and find

$$\frac{d}{dx} \sin(\underbrace{x^2 + x}_{\text{inside}}) = \cos(\underbrace{x^2 + x}_{\text{inside left alone}}) \cdot \underbrace{(2x + 1)}_{\text{derivative of the inside}}.$$

Repeated Use of the Chain Rule

We sometimes have to use the Chain Rule two or more times to find a derivative.

EXAMPLE 4 Find the derivative of $g(t) = \tan(5 - \sin 2t)$.

Solution Notice here that the tangent is a function of $5 - \sin 2t$, whereas the sine is a function of $2t$, which is itself a function of t . Therefore, by the Chain Rule,

$$\begin{aligned}
 g'(t) &= \frac{d}{dt}(\tan(5 - \sin 2t)) && \text{Derivative of } \tan u \text{ with } u = 5 - \sin 2t \\
 &= \sec^2(5 - \sin 2t) \cdot \frac{d}{dt}(5 - \sin 2t) && \text{Derivative of } 5 - \sin u \text{ with } u = 2t \\
 &= \sec^2(5 - \sin 2t) \cdot \left(0 - \cos 2t \cdot \frac{d}{dt}(2t)\right) \\
 &= \sec^2(5 - \sin 2t) \cdot (-\cos 2t) \cdot 2 \\
 &= -2(\cos 2t) \sec^2(5 - \sin 2t).
 \end{aligned}$$

The Chain Rule with Powers of a Function

If f is a differentiable function of u and if u is a differentiable function of x , then substituting $y = f(u)$ into the Chain Rule formula

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

leads to the formula

$$\frac{d}{dx} f(u) = f'(u) \frac{du}{dx}.$$

If n is any real number and f is a power function, $f(u) = u^n$, the Power Rule tells us that $f'(u) = nu^{n-1}$. If u is a differentiable function of x , then we can use the Chain Rule to extend this to the **Power Chain Rule**:

$$\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}. \quad \frac{d}{du}(u^n) = nu^{n-1}$$

EXAMPLE 5 The Power Chain Rule simplifies computing the derivative of a power of an expression.

$$\begin{aligned}
 \text{(a)} \quad \frac{d}{dx}(5x^3 - x^4)^7 &= 7(5x^3 - x^4)^6 \frac{d}{dx}(5x^3 - x^4) && \text{Power Chain Rule with } u = 5x^3 - x^4, n = 7 \\
 &= 7(5x^3 - x^4)^6(5 \cdot 3x^2 - 4x^3) \\
 &= 7(5x^3 - x^4)^6(15x^2 - 4x^3) \\
 \text{(b)} \quad \frac{d}{dx}\left(\frac{1}{3x - 2}\right) &= \frac{d}{dx}(3x - 2)^{-1} \\
 &= -1(3x - 2)^{-2} \frac{d}{dx}(3x - 2) && \text{Power Chain Rule with } u = 3x - 2, n = -1 \\
 &= -1(3x - 2)^{-2}(3) \\
 &= -\frac{3}{(3x - 2)^2}
 \end{aligned}$$

In part (b) we could also find the derivative with the Derivative Quotient Rule.

$$\begin{aligned}
 \text{(c)} \quad \frac{d}{dx}(\sin^5 x) &= 5 \sin^4 x \cdot \frac{d}{dx} \sin x && \text{Power Chain Rule with } u = \sin x, n = 5, \\
 &= 5 \sin^4 x \cos x && \text{because } \sin^n x \text{ means } (\sin x)^n, n \neq -1.
 \end{aligned}$$

HISTORICAL BIOGRAPHY

Johann Bernoulli
(1667–1748)

**Derivative of the
Absolute Value Function**

$$\frac{d}{dx}(|x|) = \frac{x}{|x|}, \quad x \neq 0$$

EXAMPLE 6 In Section 3.2, we saw that the absolute value function $y = |x|$ is not differentiable at $x = 0$. However, the function *is* differentiable at all other real numbers as we now show. Since $|x| = \sqrt{x^2}$, we can derive the following formula:

$$\begin{aligned} \frac{d}{dx}(|x|) &= \frac{d}{dx}\sqrt{x^2} \\ &= \frac{1}{2\sqrt{x^2}} \cdot \frac{d}{dx}(x^2) && \text{Power Chain Rule with } u = x^2, n = 1/2, x \neq 0 \\ &= \frac{1}{2|x|} \cdot 2x && \sqrt{x^2} = |x| \\ &= \frac{x}{|x|}, \quad x \neq 0. \end{aligned}$$

EXAMPLE 7 Show that the slope of every line tangent to the curve $y = 1/(1 - 2x)^3$ is positive.

Solution We find the derivative:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(1 - 2x)^{-3} \\ &= -3(1 - 2x)^{-4} \cdot \frac{d}{dx}(1 - 2x) && \text{Power Chain Rule with } u = (1 - 2x), n = -3 \\ &= -3(1 - 2x)^{-4} \cdot (-2) \\ &= \frac{6}{(1 - 2x)^4} \end{aligned}$$

At any point (x, y) on the curve, $x \neq 1/2$ and the slope of the tangent line is

$$\frac{dy}{dx} = \frac{6}{(1 - 2x)^4},$$

the quotient of two positive numbers. ■

EXAMPLE 8 The formulas for the derivatives of both $\sin x$ and $\cos x$ were obtained under the assumption that x is measured in radians, *not* degrees. The Chain Rule gives us new insight into the difference between the two. Since $180^\circ = \pi$ radians, $x^\circ = \pi x/180$ radians where x° is the size of the angle measured in degrees.

By the Chain Rule,

$$\frac{d}{dx}\sin(x^\circ) = \frac{d}{dx}\sin\left(\frac{\pi x}{180}\right) = \frac{\pi}{180}\cos\left(\frac{\pi x}{180}\right) = \frac{\pi}{180}\cos(x^\circ).$$

See Figure 3.25. Similarly, the derivative of $\cos(x^\circ)$ is $-(\pi/180)\sin(x^\circ)$.

The factor $\pi/180$ would compound with repeated differentiation. We see here the advantage for the use of radian measure in computations. ■

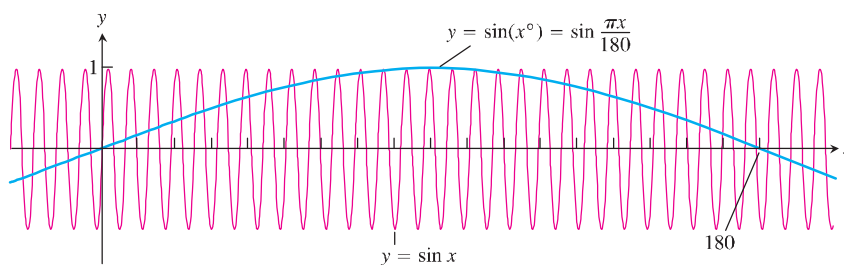


FIGURE 3.25 $\sin(x^\circ)$ oscillates only $\pi/180$ times as often as $\sin x$ oscillates. Its maximum slope is $\pi/180$ at $x = 0$ (Example 8).

Exercises 3.6

Derivative Calculations

In Exercises 1–8, given $y = f(u)$ and $u = g(x)$, find $dy/dx = f'(g(x))g'(x)$.

1. $y = 6u - 9$, $u = (1/2)x^4$
2. $y = 2u^3$, $u = 8x - 1$
3. $y = \sin u$, $u = 3x + 1$
4. $y = \cos u$, $u = -x/3$
5. $y = \cos u$, $u = \sin x$
6. $y = \sin u$, $u = x - \cos x$
7. $y = \tan u$, $u = 10x - 5$
8. $y = -\sec u$, $u = x^2 + 7x$

In Exercises 9–18, write the function in the form $y = f(u)$ and $u = g(x)$. Then find dy/dx as a function of x .

9. $y = (2x + 1)^5$
10. $y = (4 - 3x)^9$
11. $y = \left(1 - \frac{x}{7}\right)^{-7}$
12. $y = \left(\frac{x}{2} - 1\right)^{-10}$
13. $y = \left(\frac{x^2}{8} + x - \frac{1}{x}\right)^4$
14. $y = \sqrt{3x^2 - 4x + 6}$
15. $y = \sec(\tan x)$
16. $y = \cot\left(\pi - \frac{1}{x}\right)$
17. $y = \sin^3 x$
18. $y = 5 \cos^{-4} x$

Find the derivatives of the functions in Exercises 19–40.

19. $p = \sqrt{3 - t}$
20. $q = \sqrt[3]{2r - r^2}$
21. $s = \frac{4}{3\pi} \sin 3t + \frac{4}{5\pi} \cos 5t$
22. $s = \sin\left(\frac{3\pi t}{2}\right) + \cos\left(\frac{3\pi t}{2}\right)$
23. $r = (\csc \theta + \cot \theta)^{-1}$
24. $r = 6(\sec \theta - \tan \theta)^{3/2}$
25. $y = x^2 \sin^4 x + x \cos^{-2} x$
26. $y = \frac{1}{x} \sin^{-5} x - \frac{x}{3} \cos^3 x$
27. $y = \frac{1}{21} (3x - 2)^7 + \left(4 - \frac{1}{2x^2}\right)^{-1}$
28. $y = (5 - 2x)^{-3} + \frac{1}{8} \left(\frac{2}{x} + 1\right)^4$
29. $y = (4x + 3)^4 (x + 1)^{-3}$
30. $y = (2x - 5)^{-1} (x^2 - 5x)^6$
31. $h(x) = x \tan(2\sqrt{x}) + 7$
32. $k(x) = x^2 \sec\left(\frac{1}{x}\right)$
33. $f(x) = \sqrt{7 + x \sec x}$
34. $g(x) = \frac{\tan 3x}{(x + 7)^4}$
35. $f(\theta) = \left(\frac{\sin \theta}{1 + \cos \theta}\right)^2$
36. $g(t) = \left(\frac{1 + \sin 3t}{3 - 2t}\right)^{-1}$
37. $r = \sin(\theta^2) \cos(2\theta)$
38. $r = \sec \sqrt{\theta} \tan\left(\frac{1}{\theta}\right)$
39. $q = \sin\left(\frac{t}{\sqrt{t+1}}\right)$
40. $q = \cot\left(\frac{\sin t}{t}\right)$

In Exercises 41–58, find dy/dt .

41. $y = \sin^2(\pi t - 2)$
42. $y = \sec^2 \pi t$
43. $y = (1 + \cos 2t)^{-4}$
44. $y = (1 + \cot(t/2))^{-2}$
45. $y = (t \tan t)^{10}$
46. $y = (t^{-3/4} \sin t)^{4/3}$
47. $y = \left(\frac{t^2}{t^3 - 4t}\right)^3$
48. $y = \left(\frac{3t - 4}{5t + 2}\right)^{-5}$

49. $y = \sin(\cos(2t - 5))$
50. $y = \cos\left(5 \sin\left(\frac{t}{3}\right)\right)$
51. $y = \left(1 + \tan^4\left(\frac{t}{12}\right)\right)^3$
52. $y = \frac{1}{6} (1 + \cos^2(7t))^3$
53. $y = \sqrt{1 + \cos(t^2)}$
54. $y = 4 \sin(\sqrt{1 + \sqrt{t}})$
55. $y = \tan^2(\sin^3 t)$
56. $y = \cos^4(\sec^2 3t)$
57. $y = 3t(2t^2 - 5)^4$
58. $y = \sqrt{3t + \sqrt{2 + \sqrt{1 - t}}}$

Second Derivatives

Find y'' in Exercises 59–64.

59. $y = \left(1 + \frac{1}{x}\right)^3$
60. $y = (1 - \sqrt{x})^{-1}$
61. $y = \frac{1}{9} \cot(3x - 1)$
62. $y = 9 \tan\left(\frac{x}{3}\right)$
63. $y = x(2x + 1)^4$
64. $y = x^2(x^3 - 1)^5$

Finding Derivative Values

In Exercises 65–70, find the value of $(f \circ g)'$ at the given value of x .

65. $f(u) = u^5 + 1$, $u = g(x) = \sqrt{x}$, $x = 1$
66. $f(u) = 1 - \frac{1}{u}$, $u = g(x) = \frac{1}{1 - x}$, $x = -1$
67. $f(u) = \cot \frac{\pi u}{10}$, $u = g(x) = 5\sqrt{x}$, $x = 1$
68. $f(u) = u + \frac{1}{\cos^2 u}$, $u = g(x) = \pi x$, $x = 1/4$
69. $f(u) = \frac{2u}{u^2 + 1}$, $u = g(x) = 10x^2 + x + 1$, $x = 0$
70. $f(u) = \left(\frac{u - 1}{u + 1}\right)^2$, $u = g(x) = \frac{1}{x^2} - 1$, $x = -1$
71. Assume that $f'(3) = -1$, $g'(2) = 5$, $g(2) = 3$, and $y = f(g(x))$. What is y' at $x = 2$?
72. If $r = \sin(f(t))$, $f(0) = \pi/3$, and $f'(0) = 4$, then what is dr/dt at $t = 0$?
73. Suppose that functions f and g and their derivatives with respect to x have the following values at $x = 2$ and $x = 3$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	8	2	1/3	-3
3	3	-4	2π	5

Find the derivatives with respect to x of the following combinations at the given value of x .

- a. $2f(x)$, $x = 2$
- b. $f(x) + g(x)$, $x = 3$
- c. $f(x) \cdot g(x)$, $x = 3$
- d. $f(x)/g(x)$, $x = 2$
- e. $f(g(x))$, $x = 2$
- f. $\sqrt{f(x)}$, $x = 2$
- g. $1/g^2(x)$, $x = 3$
- h. $\sqrt{f^2(x) + g^2(x)}$, $x = 2$

74. Suppose that the functions f and g and their derivatives with respect to x have the following values at $x = 0$ and $x = 1$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	1	1	5	$1/3$
1	3	-4	$-1/3$	$-8/3$

Find the derivatives with respect to x of the following combinations at the given value of x .

- a. $5f(x) - g(x)$, $x = 1$ b. $f(x)g^3(x)$, $x = 0$
 c. $\frac{f(x)}{g(x) + 1}$, $x = 1$ d. $f(g(x))$, $x = 0$
 e. $g(f(x))$, $x = 0$ f. $(x^{11} + f(x))^{-2}$, $x = 1$
 g. $f(x + g(x))$, $x = 0$
75. Find ds/dt when $\theta = 3\pi/2$ if $s = \cos \theta$ and $d\theta/dt = 5$.
 76. Find dy/dt when $x = 1$ if $y = x^2 + 7x - 5$ and $dx/dt = 1/3$.

Theory and Examples

What happens if you can write a function as a composite in different ways? Do you get the same derivative each time? The Chain Rule says you should. Try it with the functions in Exercises 77 and 78.

77. Find dy/dx if $y = x$ by using the Chain Rule with y as a composite of
 a. $y = (u/5) + 7$ and $u = 5x - 35$
 b. $y = 1 + (1/u)$ and $u = 1/(x - 1)$.
78. Find dy/dx if $y = x^{3/2}$ by using the Chain Rule with y as a composite of
 a. $y = u^3$ and $u = \sqrt{x}$
 b. $y = \sqrt{u}$ and $u = x^3$.
79. Find the tangent to $y = ((x - 1)/(x + 1))^2$ at $x = 0$.
 80. Find the tangent to $y = \sqrt{x^2 - x + 7}$ at $x = 2$.
 81. a. Find the tangent to the curve $y = 2 \tan(\pi x/4)$ at $x = 1$.
 b. **Slopes on a tangent curve** What is the smallest value the slope of the curve can ever have on the interval $-2 < x < 2$? Give reasons for your answer.
82. **Slopes on sine curves**
 a. Find equations for the tangents to the curves $y = \sin 2x$ and $y = -\sin(x/2)$ at the origin. Is there anything special about how the tangents are related? Give reasons for your answer.
 b. Can anything be said about the tangents to the curves $y = \sin mx$ and $y = -\sin(x/m)$ at the origin (m a constant $\neq 0$)? Give reasons for your answer.
 c. For a given m , what are the largest values the slopes of the curves $y = \sin mx$ and $y = -\sin(x/m)$ can ever have? Give reasons for your answer.
 d. The function $y = \sin x$ completes one period on the interval $[0, 2\pi]$, the function $y = \sin 2x$ completes two periods, the function $y = \sin(x/2)$ completes half a period, and so on. Is there any relation between the number of periods $y = \sin mx$ completes on $[0, 2\pi]$ and the slope of the curve $y = \sin mx$ at the origin? Give reasons for your answer.

83. **Running machinery too fast** Suppose that a piston is moving straight up and down and that its position at time t sec is

$$s = A \cos(2\pi bt),$$

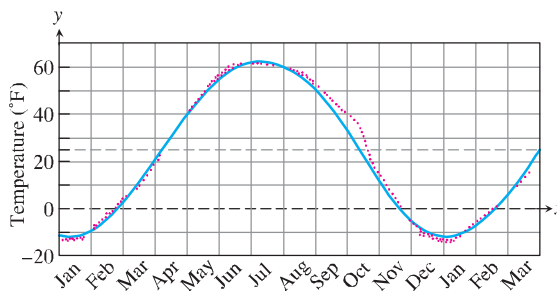
with A and b positive. The value of A is the amplitude of the motion, and b is the frequency (number of times the piston moves up and down each second). What effect does doubling the frequency have on the piston's velocity, acceleration, and jerk? (Once you find out, you will know why some machinery breaks when you run it too fast.)

84. **Temperatures in Fairbanks, Alaska** The graph in the accompanying figure shows the average Fahrenheit temperature in Fairbanks, Alaska, during a typical 365-day year. The equation that approximates the temperature on day x is

$$y = 37 \sin \left[\frac{2\pi}{365} (x - 101) \right] + 25$$

and is graphed in the accompanying figure.

- a. On what day is the temperature increasing the fastest?
 b. About how many degrees per day is the temperature increasing when it is increasing at its fastest?



85. **Particle motion** The position of a particle moving along a coordinate line is $s = \sqrt{1 + 4t}$, with s in meters and t in seconds. Find the particle's velocity and acceleration at $t = 6$ sec.
86. **Constant acceleration** Suppose that the velocity of a falling body is $v = k\sqrt{s}$ m/sec (k a constant) at the instant the body has fallen s m from its starting point. Show that the body's acceleration is constant.
87. **Falling meteorite** The velocity of a heavy meteorite entering Earth's atmosphere is inversely proportional to $\sqrt{s}$ when it is s km from Earth's center. Show that the meteorite's acceleration is inversely proportional to s^2 .
88. **Particle acceleration** A particle moves along the x -axis with velocity $dx/dt = f(x)$. Show that the particle's acceleration is $f(x)f'(x)$.
89. **Temperature and the period of a pendulum** For oscillations of small amplitude (short swings), we may safely model the relationship between the period T and the length L of a simple pendulum with the equation

$$T = 2\pi\sqrt{\frac{L}{g}},$$

where g is the constant acceleration of gravity at the pendulum's location. If we measure g in centimeters per second squared, we measure L in centimeters and T in seconds. If the pendulum is

made of metal, its length will vary with temperature, either increasing or decreasing at a rate that is roughly proportional to L . In symbols, with u being temperature and k the proportionality constant,

$$\frac{dL}{du} = kL.$$

Assuming this to be the case, show that the rate at which the period changes with respect to temperature is $kT/2$.

- 90. Chain Rule** Suppose that $f(x) = x^2$ and $g(x) = \lfloor x \rfloor$. Then the composites

$$(f \circ g)(x) = \lfloor x \rfloor^2 = x^2 \quad \text{and} \quad (g \circ f)(x) = \lfloor x^2 \rfloor = x^2$$

are both differentiable at $x = 0$ even though g itself is not differentiable at $x = 0$. Does this contradict the Chain Rule? Explain.

- T 91. The derivative of $\sin 2x$** Graph the function $y = 2 \cos 2x$ for $-2 \leq x \leq 3.5$. Then, on the same screen, graph

$$y = \frac{\sin 2(x+h) - \sin 2x}{h}$$

for $h = 1.0, 0.5$, and 0.2 . Experiment with other values of h , including negative values. What do you see happening as $h \rightarrow 0$? Explain this behavior.

- T 92. The derivative of $\cos(x^2)$** Graph $y = -2x \sin(x^2)$ for $-2 \leq x \leq 3$. Then, on the same screen, graph

$$y = \frac{\cos((x+h)^2) - \cos(x^2)}{h}$$

for $h = 1.0, 0.7$, and 0.3 . Experiment with other values of h . What do you see happening as $h \rightarrow 0$? Explain this behavior.

COMPUTER EXPLORATIONS

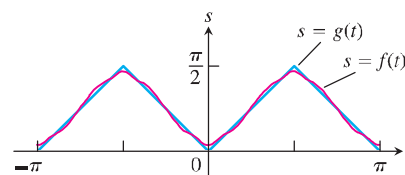
Trigonometric Polynomials

- 93.** As the accompanying figure shows, the trigonometric “polynomial”

$$\begin{aligned} s = f(t) &= 0.78540 - 0.63662 \cos 2t - 0.07074 \cos 6t \\ &\quad - 0.02546 \cos 10t - 0.01299 \cos 14t \end{aligned}$$

gives a good approximation of the sawtooth function $s = g(t)$ on the interval $[-\pi, \pi]$. How well does the derivative of f approximate the derivative of g at the points where dg/dt is defined? To find out, carry out the following steps.

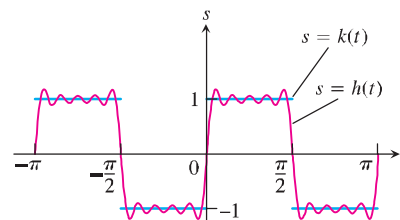
- Graph dg/dt (where defined) over $[-\pi, \pi]$.
- Find df/dt .
- Graph df/dt . Where does the approximation of dg/dt by df/dt seem to be best? Least good? Approximations by trigonometric polynomials are important in the theories of heat and oscillation, but we must not expect too much of them, as we see in the next exercise.



- 94. (Continuation of Exercise 93.)** In Exercise 93, the trigonometric polynomial $f(t)$ that approximated the sawtooth function $g(t)$ on $[-\pi, \pi]$ had a derivative that approximated the derivative of the sawtooth function. It is possible, however, for a trigonometric polynomial to approximate a function in a reasonable way without its derivative approximating the function's derivative at all well. As a case in point, the “polynomial”

$$\begin{aligned} s = h(t) &= 1.2732 \sin 2t + 0.4244 \sin 6t + 0.25465 \sin 10t \\ &\quad + 0.18189 \sin 14t + 0.14147 \sin 18t \end{aligned}$$

graphed in the accompanying figure approximates the step function $s = k(t)$ shown there. Yet the derivative of h is nothing like the derivative of k .



- Graph dk/dt (where defined) over $[-\pi, \pi]$.
- Find dh/dt .
- Graph dh/dt to see how badly the graph fits the graph of dk/dt . Comment on what you see.

3.7 Implicit Differentiation

Most of the functions we have dealt with so far have been described by an equation of the form $y = f(x)$ that expresses y explicitly in terms of the variable x . We have learned rules for differentiating functions defined in this way. Another situation occurs when we encounter equations like

$$x^3 + y^3 - 9xy = 0, \quad y^2 - x = 0, \quad \text{or} \quad x^2 + y^2 - 25 = 0.$$

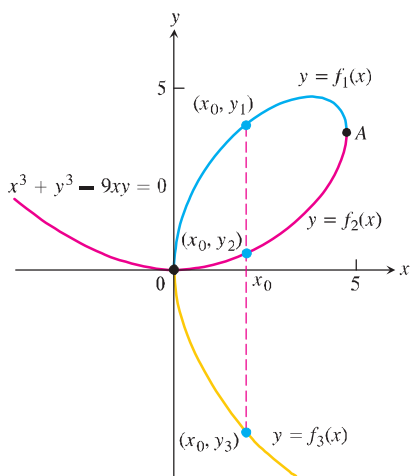


FIGURE 3.26 The curve $x^3 + y^3 - 9xy = 0$ is not the graph of any one function of x . The curve can, however, be divided into separate arcs that are the graphs of functions of x . This particular curve, called a *folium*, dates to Descartes in 1638.

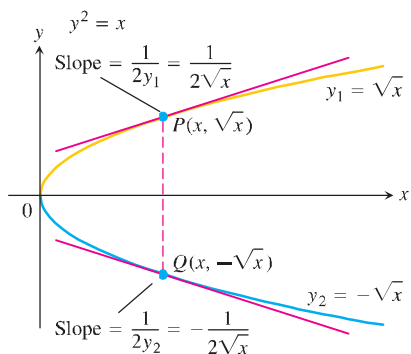


FIGURE 3.27 The equation $y^2 - x = 0$, or $y^2 = x$ as it is usually written, defines two differentiable functions of x on the interval $x > 0$. Example 1 shows how to find the derivatives of these functions without solving the equation $y^2 = x$ for y .

(See Figures 3.26, 3.27, and 3.28.) These equations define an *implicit* relation between the variables x and y . In some cases we may be able to solve such an equation for y as an explicit function (or even several functions) of x . When we cannot put an equation $F(x, y) = 0$ in the form $y = f(x)$ to differentiate it in the usual way, we may still be able to find dy/dx by *implicit differentiation*. This section describes the technique.

Implicitly Defined Functions

We begin with examples involving familiar equations that we can solve for y as a function of x to calculate dy/dx in the usual way. Then we differentiate the equations implicitly, and find the derivative to compare the two methods. Following the examples, we summarize the steps involved in the new method. In the examples and exercises, it is always assumed that the given equation determines y implicitly as a differentiable function of x so that dy/dx exists.

EXAMPLE 1 Find dy/dx if $y^2 = x$.

Solution The equation $y^2 = x$ defines two differentiable functions of x that we can actually find, namely $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$ (Figure 3.27). We know how to calculate the derivative of each of these for $x > 0$:

$$\frac{dy_1}{dx} = \frac{1}{2\sqrt{x}} \quad \text{and} \quad \frac{dy_2}{dx} = -\frac{1}{2\sqrt{x}}.$$

But suppose that we knew only that the equation $y^2 = x$ defined y as one or more differentiable functions of x for $x > 0$ without knowing exactly what these functions were. Could we still find dy/dx ?

The answer is yes. To find dy/dx , we simply differentiate both sides of the equation $y^2 = x$ with respect to x , treating $y = f(x)$ as a differentiable function of x :

$$\begin{aligned} y^2 &= x & \text{The Chain Rule gives } \frac{d}{dx}(y^2) &= \\ 2y \frac{dy}{dx} &= 1 & \frac{d}{dx}[f(x)]^2 = 2f(x)f'(x) = 2y \frac{dy}{dx}. \\ \frac{dy}{dx} &= \frac{1}{2y}. \end{aligned}$$

This one formula gives the derivatives we calculated for *both* explicit solutions $y_1 = \sqrt{x}$ and $y_2 = -\sqrt{x}$:

$$\frac{dy_1}{dx} = \frac{1}{2y_1} = \frac{1}{2\sqrt{x}} \quad \text{and} \quad \frac{dy_2}{dx} = \frac{1}{2y_2} = \frac{1}{2(-\sqrt{x})} = -\frac{1}{2\sqrt{x}}. \quad \blacksquare$$

EXAMPLE 2 Find the slope of the circle $x^2 + y^2 = 25$ at the point $(3, -4)$.

Solution The circle is not the graph of a single function of x . Rather it is the combined graphs of two differentiable functions, $y_1 = \sqrt{25 - x^2}$ and $y_2 = -\sqrt{25 - x^2}$ (Figure 3.28). The point $(3, -4)$ lies on the graph of y_2 , so we can find the slope by calculating the derivative directly, using the Power Chain Rule:

$$\left. \frac{dy_2}{dx} \right|_{x=3} = -\left. \frac{-2x}{2\sqrt{25 - x^2}} \right|_{x=3} = -\frac{-6}{2\sqrt{25 - 9}} = \frac{3}{4}. \quad \frac{d}{dx} = (25 - x^2)^{1/2} = -\frac{1}{2}(25 - x^2)^{-1/2}(-2x)$$

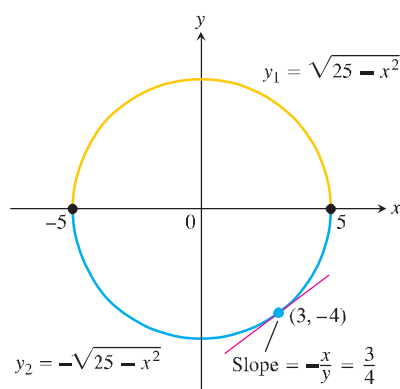


FIGURE 3.28 The circle combines the graphs of two functions. The graph of y_2 is the lower semicircle and passes through $(3, -4)$.

We can solve this problem more easily by differentiating the given equation of the circle implicitly with respect to x :

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}.$$

The slope at $(3, -4)$ is $-\frac{x}{y}\bigg|_{(3, -4)} = -\frac{3}{-4} = \frac{3}{4}$.

Notice that unlike the slope formula for dy_2/dx , which applies only to points below the x -axis, the formula $dy/dx = -x/y$ applies everywhere the circle has a slope. Notice also that the derivative involves *both* variables x and y , not just the independent variable x . ■

To calculate the derivatives of other implicitly defined functions, we proceed as in Examples 1 and 2: We treat y as a differentiable implicit function of x and apply the usual rules to differentiate both sides of the defining equation.

Implicit Differentiation

1. Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .
2. Collect the terms with dy/dx on one side of the equation and solve for dy/dx .

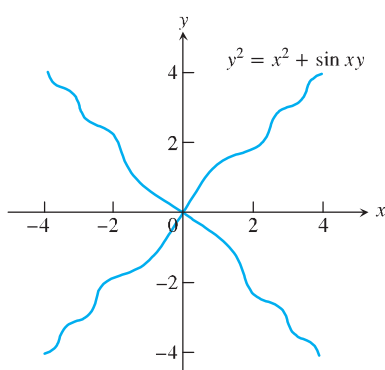


FIGURE 3.29 The graph of $y^2 = x^2 + \sin xy$ in Example 3.

EXAMPLE 3 Find dy/dx if $y^2 = x^2 + \sin xy$ (Figure 3.29).

Solution We differentiate the equation implicitly.

$$y^2 = x^2 + \sin xy$$

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(x^2) + \frac{d}{dx}(\sin xy)$$

Differentiate both sides with respect to x ...

$$2y \frac{dy}{dx} = 2x + (\cos xy) \frac{d}{dx}(xy)$$

... treating y as a function of x and using the Chain Rule.

$$2y \frac{dy}{dx} = 2x + (\cos xy) \left(y + x \frac{dy}{dx} \right)$$

Treat xy as a product.

$$2y \frac{dy}{dx} - (\cos xy) \left(x \frac{dy}{dx} \right) = 2x + (\cos xy)y$$

Collect terms with dy/dx .

$$(2y - x \cos xy) \frac{dy}{dx} = 2x + y \cos xy$$

$$\frac{dy}{dx} = \frac{2x + y \cos xy}{2y - x \cos xy}$$

Solve for dy/dx .

Notice that the formula for dy/dx applies everywhere that the implicitly defined curve has a slope. Notice again that the derivative involves *both* variables x and y , not just the independent variable x . ■

Derivatives of Higher Order

Implicit differentiation can also be used to find higher derivatives.

EXAMPLE 4 Find d^2y/dx^2 if $2x^3 - 3y^2 = 8$.

Solution To start, we differentiate both sides of the equation with respect to x in order to find $y' = dy/dx$.

$$\begin{aligned}\frac{d}{dx}(2x^3 - 3y^2) &= \frac{d}{dx}(8) \\ 6x^2 - 6yy' &= 0 && \text{Treat } y \text{ as a function of } x. \\ y' &= \frac{x^2}{y}, \quad \text{when } y \neq 0 && \text{Solve for } y'.\end{aligned}$$

We now apply the Quotient Rule to find y'' .

$$y'' = \frac{d}{dx}\left(\frac{x^2}{y}\right) = \frac{2xy - x^2y'}{y^2} = \frac{2x}{y} - \frac{x^2}{y^2} \cdot y'$$

Finally, we substitute $y' = x^2/y$ to express y'' in terms of x and y .

$$y'' = \frac{2x}{y} - \frac{x^2}{y^2} \left(\frac{x^2}{y}\right) = \frac{2x}{y} - \frac{x^4}{y^3}, \quad \text{when } y \neq 0$$

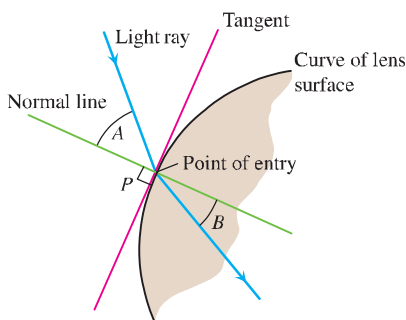


FIGURE 3.30 The profile of a lens, showing the bending (refraction) of a ray of light as it passes through the lens surface.

Lenses, Tangents, and Normal Lines

In the law that describes how light changes direction as it enters a lens, the important angles are the angles the light makes with the line perpendicular to the surface of the lens at the point of entry (angles A and B in Figure 3.30). This line is called the *normal* to the surface at the point of entry. In a profile view of a lens like the one in Figure 3.30, the **normal** is the line perpendicular to the tangent of the profile curve at the point of entry.

EXAMPLE 5 Show that the point $(2, 4)$ lies on the curve $x^3 + y^3 - 9xy = 0$. Then find the tangent and normal to the curve there (Figure 3.31).

Solution The point $(2, 4)$ lies on the curve because its coordinates satisfy the equation given for the curve: $2^3 + 4^3 - 9(2)(4) = 8 + 64 - 72 = 0$.

To find the slope of the curve at $(2, 4)$, we first use implicit differentiation to find a formula for dy/dx :

$$\begin{aligned}x^3 + y^3 - 9xy &= 0 \\ \frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) - \frac{d}{dx}(9xy) &= \frac{d}{dx}(0) \\ 3x^2 + 3y^2 \frac{dy}{dx} - 9\left(x \frac{dy}{dx} + y \frac{dx}{dx}\right) &= 0 && \text{Differentiate both sides with respect to } x. \\ (3y^2 - 9x) \frac{dy}{dx} + 3x^2 - 9y &= 0 && \text{Treat } xy \text{ as a product and } y \text{ as a function of } x. \\ 3(y^2 - 3x) \frac{dy}{dx} &= 9y - 3x^2 \\ \frac{dy}{dx} &= \frac{3y - x^2}{y^2 - 3x} && \text{Solve for } dy/dx.\end{aligned}$$

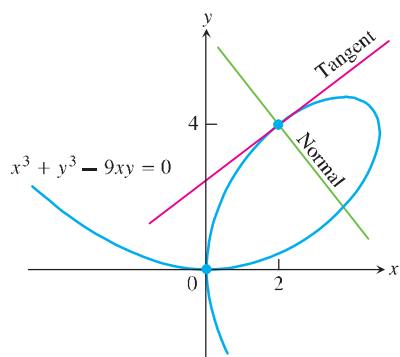


FIGURE 3.31 Example 5 shows how to find equations for the tangent and normal to the folium of Descartes at $(2, 4)$.

We then evaluate the derivative at $(x, y) = (2, 4)$:

$$\left. \frac{dy}{dx} \right|_{(2,4)} = \left. \frac{3y - x^2}{y^2 - 3x} \right|_{(2,4)} = \frac{3(4) - 2^2}{4^2 - 3(2)} = \frac{8}{10} = \frac{4}{5}.$$

The tangent at $(2, 4)$ is the line through $(2, 4)$ with slope $4/5$:

$$\begin{aligned} y &= 4 + \frac{4}{5}(x - 2) \\ y &= \frac{4}{5}x + \frac{12}{5}. \end{aligned}$$

The normal to the curve at $(2, 4)$ is the line perpendicular to the tangent there, the line through $(2, 4)$ with slope $-5/4$:

$$\begin{aligned} y &= 4 - \frac{5}{4}(x - 2) \\ y &= -\frac{5}{4}x + \frac{13}{2}. \end{aligned}$$

The quadratic formula enables us to solve a second-degree equation like $y^2 - 2xy + 3x^2 = 0$ for y in terms of x . There is a formula for the three roots of a cubic equation that is like the quadratic formula but much more complicated. If this formula is used to solve the equation $x^3 + y^3 = 9xy$ in Example 5 for y in terms of x , then three functions determined by the equation are

$$y = f(x) = \sqrt[3]{-\frac{x^3}{2} + \sqrt{\frac{x^6}{4} - 27x^3}} + \sqrt[3]{-\frac{x^3}{2} - \sqrt{\frac{x^6}{4} - 27x^3}}$$

and

$$y = \frac{1}{2} \left[-f(x) \pm \sqrt{-3} \left(\sqrt[3]{-\frac{x^3}{2} + \sqrt{\frac{x^6}{4} - 27x^3}} - \sqrt[3]{-\frac{x^3}{2} - \sqrt{\frac{x^6}{4} - 27x^3}} \right) \right].$$

Using implicit differentiation in Example 5 was much simpler than calculating dy/dx directly from any of the above formulas. Finding slopes on curves defined by higher-degree equations usually requires implicit differentiation.

Exercise 3.7

Differentiating Implicitly

Use implicit differentiation to find dy/dx in Exercises 1–14.

- $x^2y + xy^2 = 6$
- $x^3 + y^3 = 18xy$
- $2xy - y^2 = x + y$
- $x^3 - xy + y^3 = 1$
- $x^2(x - y)^2 = x^2 - y^2$
- $(3xy + 7)^2 = 6y$
- $y^2 = \frac{x-1}{x+1}$
- $x^3 = \frac{2x-y}{x+3y}$
- $x = \tan y$
- $xy = \cot(xy)$
- $x + \tan(xy) = 0$
- $x^4 + \sin y = x^3y^2$
- $y \sin\left(\frac{1}{y}\right) = 1 - xy$
- $x \cos(2x + 3y) = y \sin x$

Find $dr/d\theta$ in Exercises 15–18.

- $\theta^{1/2} + r^{1/2} = 1$
- $r - 2\sqrt{\theta} = \frac{3}{2}\theta^{2/3} + \frac{4}{3}\theta^{3/4}$
- $\sin(r\theta) = \frac{1}{2}$
- $\cos r + \cot \theta = r\theta$

Second Derivatives

In Exercises 19–26, use implicit differentiation to find dy/dx and then d^2y/dx^2 .

- $x^2 + y^2 = 1$
- $x^{2/3} + y^{2/3} = 1$
- $y^2 = x^2 + 2x$
- $y^2 - 2x = 1 - 2y$
- $2\sqrt{y} = x - y$
- $xy + y^2 = 1$

25. If $x^3 + y^3 = 16$, find the value of d^2y/dx^2 at the point $(2, 2)$.
 26. If $xy + y^2 = 1$, find the value of d^2y/dx^2 at the point $(0, -1)$.

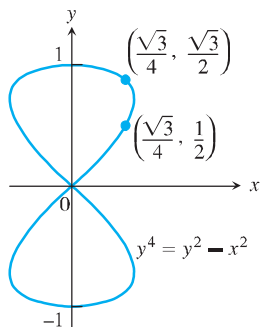
In Exercises 27 and 28, find the slope of the curve at the given points.

27. $y^2 + x^2 = y^4 - 2x$ at $(-2, 1)$ and $(-2, -1)$
 28. $(x^2 + y^2)^2 = (x - y)^2$ at $(1, 0)$ and $(1, -1)$

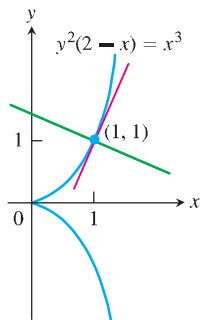
Slopes, Tangents, and Normals

In Exercises 29–38, verify that the given point is on the curve and find the lines that are (a) tangent and (b) normal to the curve at the given point.

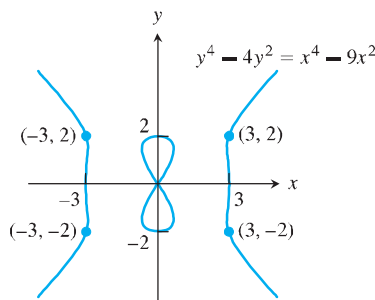
29. $x^2 + xy - y^2 = 1$, $(2, 3)$
 30. $x^2 + y^2 = 25$, $(3, -4)$
 31. $x^2y^2 = 9$, $(-1, 3)$
 32. $y^2 - 2x - 4y - 1 = 0$, $(-2, 1)$
 33. $6x^2 + 3xy + 2y^2 + 17y - 6 = 0$, $(-1, 0)$
 34. $x^2 - \sqrt{3}xy + 2y^2 = 5$, $(\sqrt{3}, 2)$
 35. $2xy + \pi \sin y = 2\pi$, $(1, \pi/2)$
 36. $x \sin 2y = y \cos 2x$, $(\pi/4, \pi/2)$
 37. $y = 2 \sin(\pi x - y)$, $(1, 0)$
 38. $x^2 \cos^2 y - \sin y = 0$, $(0, \pi)$
 39. **Parallel tangents** Find the two points where the curve $x^2 + xy + y^2 = 7$ crosses the x -axis, and show that the tangents to the curve at these points are parallel. What is the common slope of these tangents?
 40. **Normals parallel to a line** Find the normals to the curve $xy + 2x - y = 0$ that are parallel to the line $2x + y = 0$.
 41. **The eight curve** Find the slopes of the curve $y^4 = y^2 - x^2$ at the two points shown here.



42. **The cissoid of Diocles (from about 200 B.C.)** Find equations for the tangent and normal to the cissoid of Diocles $y^2(2 - x) = x^3$ at $(1, 1)$.



43. **The devil's curve (Gabriel Cramer, 1750)** Find the slopes of the devil's curve $y^4 - 4y^2 = x^4 - 9x^2$ at the four indicated points.



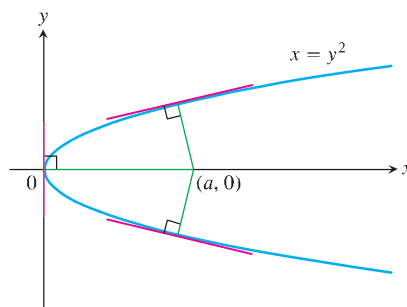
44. **The folium of Descartes** (See Figure 3.26.)
 a. Find the slope of the folium of Descartes $x^3 + y^3 - 9xy = 0$ at the points $(4, 2)$ and $(2, 4)$.
 b. At what point other than the origin does the folium have a horizontal tangent?
 c. Find the coordinates of the point A in Figure 3.26, where the folium has a vertical tangent.

Theory and Examples

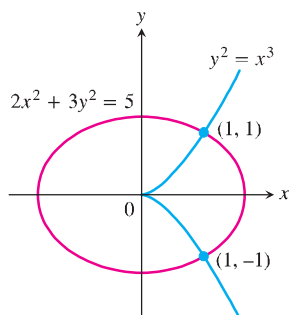
45. **Intersecting normal** The line that is normal to the curve $x^2 + 2xy - 3y^2 = 0$ at $(1, 1)$ intersects the curve at what other point?
 46. **Power rule for rational exponents** Let p and q be integers with $q > 0$. If $y = x^{p/q}$, differentiate the equivalent equation $y^q = x^p$ implicitly and show that, for $y \neq 0$,

$$\frac{d}{dx} x^{p/q} = \frac{p}{q} x^{(p/q)-1}.$$

47. **Normals to a parabola** Show that if it is possible to draw three normals from the point $(a, 0)$ to the parabola $x = y^2$ shown in the accompanying diagram, then a must be greater than $1/2$. One of the normals is the x -axis. For what value of a are the other two normals perpendicular?



48. Is there anything special about the tangents to the curves $y^2 = x^3$ and $2x^2 + 3y^2 = 5$ at the points $(1, \pm 1)$? Give reasons for your answer.

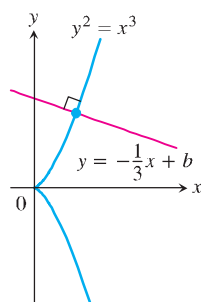


49. Verify that the following pairs of curves meet orthogonally.

a. $x^2 + y^2 = 4$, $x^2 = 3y^2$

b. $x = 1 - y^2$, $x = \frac{1}{3}y^2$

50. The graph of $y^2 = x^3$ is called a **semicubical parabola** and is shown in the accompanying figure. Determine the constant b so that the line $y = -\frac{1}{3}x + b$ meets this graph orthogonally.



T In Exercises 51 and 52, find both dy/dx (treating y as a differentiable function of x) and dx/dy (treating x as a differentiable function of y). How do dy/dx and dx/dy seem to be related? Explain the relationship geometrically in terms of the graphs.

51. $xy^3 + x^2y = 6$

52. $x^3 + y^2 = \sin^2 y$

COMPUTER EXPLORATIONS

Use a CAS to perform the following steps in Exercises 53–60.

- Plot the equation with the implicit plotter of a CAS. Check to see that the given point P satisfies the equation.
 - Using implicit differentiation, find a formula for the derivative dy/dx and evaluate it at the given point P .
 - Use the slope found in part (b) to find an equation for the tangent line to the curve at P . Then plot the implicit curve and tangent line together on a single graph.
53. $x^3 - xy + y^3 = 7$, $P(2, 1)$
54. $x^5 + y^3x + yx^2 + y^4 = 4$, $P(1, 1)$
55. $y^2 + y = \frac{2+x}{1-x}$, $P(0, 1)$
56. $y^3 + \cos xy = x^2$, $P(1, 0)$
57. $x + \tan\left(\frac{y}{x}\right) = 2$, $P\left(1, \frac{\pi}{4}\right)$
58. $xy^3 + \tan(x + y) = 1$, $P\left(\frac{\pi}{4}, 0\right)$
59. $2y^2 + (xy)^{1/3} = x^2 + 2$, $P(1, 1)$
60. $x\sqrt{1 + 2y} + y = x^2$, $P(1, 0)$

3.8

Related Rates

In this section we look at problems that ask for the rate at which some variable changes when it is known how the rate of some other related variable (or perhaps several variables) changes. The problem of finding a rate of change from other known rates of change is called a *related rates problem*.

Related Rates Equations

Suppose we are pumping air into a spherical balloon. Both the volume and radius of the balloon are increasing over time. If V is the volume and r is the radius of the balloon at an instant of time, then

$$V = \frac{4}{3}\pi r^3.$$